

Homework 3

Stat 419

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1.

- a. To assign participants, we rolled a die. Results of 1-3 assigned the patient to the control group, and results of 4-6 were assigned to the treatment. There were 13 patients assigned to the control group, and 11 assigned to the treatment group.
- b. $(24 \text{ choose } 12)(0.5)^{12}(0.5)^{12} = 0.1612 \rightarrow 16.12\%$ chance of having groups of equal size using simple randomization.
Probability of between 11-13 patients = $0.2976 + 0.1612 \rightarrow 45.88\%$
- c. *Shown in table on final page, column 5*
- d. This technique is very straight-forward and easy to implement, not to mention quick. The process could be sped up even further with the use of computer programs instead of rolling a die by hand, handling large sample sizes more efficiently. However, the method is limited because it risks having groups of unequal size, especially if the total sample size is small. A small sample makes the existence of unequal groups more likely, as the sample assignment probability converges toward the expected probability of 0.5 as $n \rightarrow \infty$

2.

- a. The table below shows the possible block configuration, using a block size of 6: 0's represent an assignment to the control group, 1's represent an assignment to the treatment group.

000111	111000
001011	110100
001101	110010
001110	110001
011100	100011
011010	100101
011001	100110
010110	101001
010011	101100
010101	101010

- b. Since the sample size is 24, we will select 4 blocks at random from the above list. This was done using a random number generator from 1-20, re-rolling any repeats. The selected blocks were:
Block 20 = 101010, Block 1 = 000111, Block 3 = 001101, Block 14 = 110001

The first 6 patients were put into a temporary “Group 1”, the next 6 “Group 2”, next 6 “Group 3”, and the final 6 “Group 4”. To allocate patients, we used a random number generator from 1-4, determining which block each group was assigned to. Of course, the range of the generator decreases with each assignment. Group 1: Block 20, Group 2: Block 3, Group 3: Block 14, Group 4: Block 1

- c. *Shown in table on final page, column 6*
- d. The major advantage of this technique is that the treatment and control are equal in size. Block randomization is also relatively simple to implement, especially with the help of computer software. However, there is a chance that either group will have a significantly different demographic to the other. That is, the method does not take into account any potential covariates. Below are three tables, one for each covariate and the number of people with said covariate in either group.

Group/Gender	Male	Female
Treatment	5	7
Control	7	5

Relatively even assignment, perhaps insignificant effect on results.

Group/Age	26-35	36-45	46-55
Treatment	3	5	4
Control	2	4	6

Mixed assignment, unequal within groups *and* between age ranges.

Group/Activity	Sedentary	Moderate	High
Treatment	5	3	4
Control	2	2	8

Very uneven groups, especially in the control group: $\frac{2}{3}$ assigned are highly active.

Major disparities between subgroup covariates can be problematic because it can lead to inaccurate results. That is, any of the covariates may have an unforeseen effect on the outcome, potentially leading to skewed results if either the treatment group or control group is not balanced with respect to these variables. For instance, a younger person who is more active is surely more likely to recover from an injury more quickly than an older, less active person. If the treatment group has a relatively high proportion of younger healthy people, a study may falsely conclude that the new treatment is extremely effective in treating achilles injuries. This could be inaccurate if the covariate was not accounted for.

3.

- a. We create a strata in this case by creating blocks using categories of gender and age. We were able to get 4 equally large blocks of 6. First here are our blocks.

Block #	Gender and Age
1	Female 26-45
2	Female 46-55
3	Male 26-38
4	Male 39-55

Now we will use block randomization, using the same table we did in problem 2a.

000111	111000
001011	110100
001101	110010
001110	110001
011100	100011
011010	100101
011001	100110
010110	101001
010011	101100
010101	101010

Now to randomize, we will use a random number generator to assign a block configuration to a block, with a size of 20 because that's how many options there are. We will also eliminate a configuration once it is already used so we don't reuse patterns. This will ensure that there will be an equal number in the control and treatment groups. The first value for the block category would get the first number in the block configuration. For example, in Block 3, Patient 2 of all the participants would be in the control group, because they're the first Male 26-28 in the dataset, and the first value is 1.

b.

Block #	Block Configuration (C=0, T=1)
1 (Female 26-45)	101100 (#19)

2 (Female 46-55)	001101 (#3)
3 (Male 26-40)	111000 (#11)
4 (Male 41-55)	011100 (#5)

- c. An advantage of it is that each block configuration has 3 of each treatment and control, so it ensures that the number in the treatment group and control group are equal in the end. A disadvantage is in order to make it work, we had to change the groups up a little bit, for example we had to split the 36-45 age group into the other two respective groups for it to work. Another potential disadvantage is that stratification does not work well if there are a large number of covariates.
- 4.
- a. To assign participants, we rolled a die. Results of 1-3 assigned the patient to the control group, and results of 4-6 were assigned to the treatment. The groups were populated equally, with 5 participants in each.

b.

Control Group

	Male	Female
26-35	1	0
36-45	1	1
46-55	1	1

Treatment Group

	Male	Female
26-35	1	1
36-45	0	1
46-55	0	2

c. Taves method

- i. Patient 11 is a 36-year-old female; the control group had 3+0, and the treatment had 4+1, so we put her in the control group.
 Patient 12 is a 35-year-old female; the control group had 3+1, and the treatment had 4+1, so we put her in the control group.

ii.

Control Group

	Male	Female
26-35	1	1
36-45	1	2
46-55	1	1

Treatment Group

	Male	Female
26-35	1	1
36-45	0	1
46-55	0	2

d. Pocock & Simon method

i. Patient 13: If assigned to control: 4 male - 1 male = 3; 4 middle aged - 1 middle aged = 3, $(3+3 = 6)$.

If assigned to treatment: 3 male - 2 male = 1; 3 male - 2 male = 1, $(1+1 = 2)$.

Conclusion: $2 < 6$, so patient 13 goes into the treatment group.

Patient 14: If assigned to control: 5 female - 4 female = 1; 3 old - 2 old = 1, $(1+1 = 2)$.

If assigned to treatment: 5 female - 4 female = 1; 3 old - 2 old = 1, $(1+1 = 2)$.

Conclusion: $2 = 2$, so patient 14 is assigned to the treatment group because we used the total number of patients in each group as a tiebreaker.

ii.

Control Group

	Male	Female
26-35	1	1
36-45	1	2
46-55	1	1

Treatment Group

	Male	Female
26-35	1	1
36-45	1	1
46-55	0	3

5.

- a. The advantage to simple randomization is that it is quick and easy, especially if performed by a computer. However, its drawback is that the groups will not necessarily be of the same size, and this is particularly harmful when dealing with small sample sizes. It also does not perform well with low sample sizes, as it cannot be expected for the groups to be approximately equal in size (WLLN). Block randomization ensures the groups are of equal size, but does not take into account any potential confounding variables, potentially leading to inaccurate conclusions. Stratified randomization however is useful as it *is* able to account for these covariates, but is also limited in that its effectiveness becomes limited if there is a large number of covariates. It is also more complicated to implement, potentially leading to errors, and perhaps selection bias as the researcher is specifically choosing where to assign a given participant.
- b. The Taves method is easier to compute for the user, as you do not need to compare the temporary values for the control and treatment groups when performing the calculation, whereas the Pocock and Simon method seems to be a much more complicated method of arriving at a similar conclusion. Both methods evaluate the specific covariates and assign the patient into the group with fewer people. A specific potential weakness of both methods is that they can become predictable, essentially defeating the purpose of randomization.

patient id	gender	age	lifestyle	Simple	Block	Strat.	Taves
1	female	28	sedentary	0	1	1	1
2	male	32	very active	0	0	1	0
3	male	50	mod active	0	1	0	0
4	male	44	very active	0	0	1	0
5	female	44	mod active	1	1	0	0
6	female	49	very active	1	0	0	0
7	female	42	very active	0	0	1	1
8	female	52	very active	1	0	0	1
9	male	26	very active	1	0	1	1
10	female	47	sedentary	0	1	1	1
11	female	36	mod active	1	1	1	0
12	female	35	very active	0	1	0	0
13	male	38	very active	0	0	1	-
14	female	48	mod active	0	0	1	-
15	female	38	very active	0	1	0	-
16	male	46	sedentary	0	1	1	-
17	male	37	mod active	1	0	0	-
18	male	40	very active	0	1	0	-
19	male	29	sedentary	1	1	0	-
20	male	41	sedentary	0	1	1	-
21	male	55	sedentary	1	0	0	-
22	male	49	very active	1	0	0	-
23	female	53	sedentary	1	0	0	-
24	female	52	very active	1	1	1	-